INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

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THE SITE

Location	OK facility OK_way_1387080629 -- station KJLN, America/Chicago
Building	OSM 1387080629 OSM way 1387080629
Facade gap	not applicable -- one building, no second facade
Weather record	43,684 real hourly records from KJLN
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2024-02-27 (crossing)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 5 of 24 hours with 1 mode change. That is 3 more than the reactive incumbent, at 0 unsafe hours against its 0. 4 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	5 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	2 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s) 4 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	-	20.810	21.0	20.560	0.993
01	mechanical	no	dry-bulb	21.110	21.0	21.110	0.927
02	mechanical	no	dry-bulb	21.115	21.0	21.110	0.876
03	mechanical	yes	-	20.840	21.0	20.560	0.822
04	mechanical	no	dry-bulb	21.115	21.0	20.560	0.779
05	mechanical	yes	-	20.555	21.0	19.440	0.736
06	mechanical	yes	-	20.555	21.0	19.440	0.888
07	mechanical	no	dry-bulb	21.950	21.0	20.560	1.145

08	mechanical	no	dry-bulb	23.330	21.0	21.110	1.761
09	mechanical	no	dry-bulb	24.725	21.0	21.670	1.991
10	mechanical	no	dry-bulb	26.670	21.0	23.890	1.975
11	mechanical	no	dry-bulb	27.225	21.0	26.110	1.491
12	mechanical	no	dry-bulb	27.225	21.0	26.670	1.312
13	mechanical	no	dry-bulb	28.335	21.0	27.780	1.152
14	mechanical	no	dry-bulb	28.335	21.0	26.670	0.968
15	mechanical	no	dry-bulb	27.780	21.0	26.110	0.856
16	mechanical	no	dry-bulb	27.505	21.0	25.560	0.845
17	mechanical	no	dry-bulb	25.835	21.0	23.890	1.173
18	mechanical	no	dry-bulb	24.445	21.0	22.780	1.186
19	FREE	yes	-	18.610	21.0	12.220	1.382
20	FREE	yes	-	15.830	21.0	8.330	1.554
21	FREE	yes	-	13.615	21.0	5.000	1.384
22	FREE	yes	-	7.220	21.0	2.220	1.247
23	FREE	yes	-	4.445	21.0	0.560	0.905

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.810 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.110 C against a 21.0 C limit, so it fails by 0.110 C. It would take a limit 0.110 C higher, or a bound 0.110 C tighter, to change this hour.

02:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.115 C against a 21.0 C limit, so it fails by 0.115 C. It would take a limit 0.115 C higher, or a bound 0.115 C tighter, to change this hour.

03:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.840 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.115 C against a 21.0 C limit, so it fails by 0.115 C. It would take a limit 0.115 C higher, or a bound 0.115 C tighter, to change this hour.

05:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.555 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.555 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.950 C against a 21.0 C limit, so it fails by 0.950 C. It would take a limit 0.950 C higher, or a bound 0.950 C tighter, to change this hour.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.330 C against a 21.0 C limit, so it fails by 2.330 C. It would take a limit 2.330 C higher, or a bound 2.330 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.725 C against a 21.0 C limit, so it fails by 3.725 C. It would take a limit 3.725 C higher, or a bound 3.725 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.670 C against a 21.0 C limit, so it fails by 5.670 C. It would take a limit 5.670 C higher, or a bound 5.670 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.225 C against a 21.0 C limit, so it fails by 6.225 C. It would take a limit 6.225 C higher, or a bound 6.225 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.225 C against a 21.0 C limit, so it fails by 6.225 C. It would take a limit 6.225 C higher, or a bound 6.225 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.335 C against a 21.0 C limit, so it fails by 7.335 C. It would take a limit 7.335 C higher, or a bound 7.335 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 28.335 C against a 21.0 C limit, so it fails by 7.335 C. It would take a limit 7.335 C higher, or a bound 7.335 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.780 C against a 21.0 C limit, so it fails by 6.780 C. It would take a limit 6.780 C higher, or a bound 6.780 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.505 C against a 21.0 C limit, so it fails by 6.505 C. It would take a limit 6.505 C higher, or a bound 6.505 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.835 C against a 21.0 C limit, so it fails by 4.835 C. It would take a limit 4.835 C higher, or a bound 4.835 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.445 C against a 21.0 C limit, so it fails by 3.445 C. It would take a limit 3.445 C higher, or a bound 3.445 C tighter, to change this hour.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.610 C, which is 2.390 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.382 C, and the margin is measured, not chosen: 1.382 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.830 C, which is 5.170 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.554 C, and the margin is measured, not chosen: 1.554 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.615 C, which is 7.385 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.384 C, and the margin is measured, not chosen: 1.384 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.220 C, which is 13.780 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.247 C, and the margin is measured, not chosen: 1.247 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 4.445 C, which is 16.555 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.905 C, and the margin is measured, not chosen: 0.905 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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